Trust Me or Check Me? Self-Reported Confidence, Verification, and Decision Authority in Language Models

Anonymous Author(s)
Affiliation
Address
email

Abstract

1 Introduction

Large language models are increasingly asked not only to produce answers, but also to influence whether those answers receive further scrutiny. A system may answer directly, seek additional information, invoke a tool, defer to a human, or recommend that an answer be independently checked before it is used. This creates a second-order decision problem beyond answer generation itself: *when should an already-generated answer be used as is, and when should it be verified first?* When the same AI system both generates an answer and participates in deciding whether that answer needs checking, uncertainty becomes part of a decision-authority mechanism: the system can shape not only what information is produced, but also when independent judgment enters the loop.

A natural control signal for this decision is the model's own reported confidence. Suppose a model has already produced an answer and reports a probability q that the answer is correct. If independent verification costs C, while using a wrong answer without verification incurs cost L, then treating q as the probability of correctness induces the simple expected-cost rule

$$(1-q)L > C \quad \Longrightarrow \quad \text{VERIFY\_FIRST}.$$

This apparently straightforward rule hides two different questions. First, will a model's downstream verification behavior actually follow the policy implied by its stated confidence? Second, if it does, will stronger agreement with that policy produce better decisions about which answers actually need checking? These questions are distinct. A verification policy can be highly consistent with reported confidence even when that confidence is poorly aligned with whether a particular answer is correct.

We study these questions with a controlled post-answer design that separates answer generation, confidence elicitation, and verification policy. For each question, the model first produces a multiple-choice answer, which is frozen. It then reports a probability q that this frozen answer is correct, which is also frozen. Only afterward does the model choose between using the answer without verification and verifying it first. At this final stage, we vary whether the previously reported q is hidden or explicitly shown, whether verification authority is described as belonging to a human or to the AI system, and the cost of using a wrong answer without verification. Because the answer and reported confidence are held fixed before these manipulations, the experiment isolates changes in downstream verification policy rather than changes in answer generation or confidence elicitation.

We evaluate 500 deterministic MMLU-Pro questions across four model families: OpenAI GPT-5.6 Sol, Anthropic Claude Sonnet 5, Google Gemini 3.8 Flash, and xAI Grok 4.20 non-reasoning. For each model–question pair, the final decision crosses two decision-authority conditions, two confidence-visibility conditions, and four error costs $L \in \{2, 5, 10, 20\}$, with verification cost fixed

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at $C = 1$. This yields 32,000 primary verification decisions. Statistical comparisons use the question as the independent sampling unit and question-level paired bootstrap confidence intervals.

Explicitly surfacing the model's own previously reported confidence strongly changes verification behavior, but not in a common direction across model families. For Claude Sonnet 5, visible confidence increases verification by approximately 12.8–22.4 percentage points across authority and cost conditions. For GPT-5.6 Sol, it decreases verification by 16.2–20.8 points throughout. Grok 4.20 similarly decreases verification by 8.6–16.4 points. Gemini 3.8 Flash shows smaller effects at low error costs but substantially higher verification when confidence is visible at larger costs, with increases of roughly 12–17 points at $L = 10$ and $L = 20$. Thus, the same intervention— making self-reported confidence explicit to the downstream decision—can produce large and opposite policy shifts across model families.

The most revealing case is GPT-5.6 Sol. We define *confidence-policy alignment* by comparing the model's final action with the action implied by its frozen reported confidence under the supplied cost structure. When confidence is hidden, GPT-5.6 Sol disagrees with this confidence-implied policy on roughly 19.6–27.2% of questions across authority and cost conditions. When the same reported q is explicitly shown, disagreement collapses to 0.2–2.2%. Its behavior simultaneously becomes almost perfectly ordered by the cost of error: stake-monotonicity violations fall from 95 to 1 per 1,000 trajectories.

Yet this increased alignment with reported confidence does not improve actual error filtering. At the highest error cost, among GPT-5.6 Sol answers that are in fact wrong, the fraction used without verification rises from 32.18% to 56.32% under human decision authority and from 33.33% to 54.02% under AI decision authority when confidence is made visible. In this case, explicitly surfacing confidence therefore produces a policy that is substantially more consistent with the model's stated uncertainty while leaving substantially more of its high-cost errors unchecked. Greater confidence-policy alignment and better verification quality are not the same property.

This distinction also matters when comparing models. A system can catch nearly all of its wrong answers simply by verifying almost everything, but such a policy defeats the purpose of selective verification. Conversely, reducing verification burden can leave more errors unchecked. We therefore evaluate verification behavior using both error filtering and unnecessary verification of correct answers rather than interpreting raw verification rate as a measure of quality.

Decision authority produces a smaller and less universal effect. Claude shows the clearest directional shift, generally verifying more when the AI system rather than the human is described as controlling the decision, particularly at higher error costs. The other model families show small or mixed average effects. We therefore do not find a general human-versus-AI authority asymmetry across models. Finally, on a prespecified 100-question robustness subset, the major confidence-visibility directions persist under one semantically equivalent Stage-3 paraphrase, although we do not claim invariance to arbitrary prompt wording.

These results concern model behavior rather than human responses. We do not measure whether people trust, follow, resist, or override a model's verification recommendation. Instead, we isolate a technical prerequisite for such questions: whether a model's own uncertainty can reliably govern when its outputs are subjected to independent scrutiny. This matters for human–AI systems in which an AI does more than provide information and also participates in determining when human judgment, external evidence, or additional computation should intervene.

Contributions. Our contributions are threefold. First, we introduce a controlled post-answer evaluation that freezes both answer and reported confidence before separately varying confidence visibility, decision authority, and error cost across 32,000 matched verification decisions. Second, we show that explicitly surfacing self-reported confidence can strongly reshape verification policy in opposite directions across model families, cautioning against treating reported confidence as a universally portable verification-control signal. Third, we identify a direct divergence between confidence-policy alignment and verification quality: for GPT-5.6 Sol, visible confidence makes downstream actions nearly coincide with the policy implied by reported q while substantially more wrong high-cost answers remain unverified.

2 Related Work

Self-reported uncertainty in language models. A growing literature studies whether language models can express useful uncertainty about their own answers. Early work showed that models can be trained to verbalize probabilities that are meaningfully calibrated [Lin et al., 2022], while subsequent studies examined black-box confidence elicitation in instruction-tuned and closed-source models [Tian et al., 2023, Xiong et al., 2024]. These results motivate self-reported confidence as an accessible uncertainty signal, particularly when token probabilities or model internals are unavailable. At the same time, calibration is not sufficient for our setting: even a confidence score that is informative about correctness may or may not govern a model's later decision about whether its answer should be checked.

Confidence, abstention, and downstream action. Recent work has begun to examine this gap directly. RiskEval evaluates whether LLMs adapt abstention behavior as the penalty for an incorrect answer changes, finding that models can fail to convert verbal uncertainty into appropriately risk-sensitive abstention policies [Wang et al., 2026]. Kumaran [Kumaran, 2026] uses a two-stage answer–confidence–abstention paradigm and finds that verbal confidence predicts subsequent commitment more strongly than objective correctness, distinguishing verbal confidence from token-probability-based uncertainty. More broadly, selective prediction and abstention methods use uncertainty to trade coverage for error reduction, including approaches that explicitly control the error rate among accepted answers [Dong and Shinnou, 2026].

Our experiment is complementary to this work. Rather than asking only whether reported confidence predicts a later decision, we freeze the answer and reported probability and intervene on whether that same probability is explicitly available at decision time. This permits a matched comparison of the downstream policy with confidence hidden versus visible. It also separates *confidence-policy alignment*—whether the action agrees with the decision rule implied by reported q—from *verification quality*— whether wrong answers are filtered while unnecessary verification of correct answers is avoided.

Reliance and decision authority in human–AI systems. Research on AI-assisted decision making distinguishes mere reliance from *appropriate* reliance: accepting useful AI advice while rejecting or scrutinizing erroneous advice [Schemmer et al., 2022]. Interface interventions can alter this behavior; for example, cognitive forcing mechanisms can reduce human overreliance on incorrect AI recommendations [Buçinca et al., 2021]. Our study does not measure human reliance. Instead, it examines an upstream system-design question relevant to such interactions: when an AI participates in deciding whether its own output deserves independent scrutiny, how do its reported uncertainty and the location of decision authority shape that recommendation?

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